

Team C.R.A.P

RMB – Frankenboard

What is it

RMB-Frankenboard. A mix of different pcb modules hooked up, flashed and ready to go with a brushless build. Ment to be used with a neutron solenoid ramping a 30 dps out of the box. Comes with an esp32 MCU and optic gates on the back for the feedback. Controlled via BT app on your phone and a select fire switch on the blaster for fire modes on the go.

Integrated into the newest version of Mjöltnir evolved

Features

- 2 Stage Brushless motors wired directly to RMB-Frankenboard
- Foam Projectiles, Half Darts or Long Darts you could also potentially use rival balls or disks
- Solenoid Pusher
- Optical Interrupt Gate on PCB for closed loop Pusher feed back. Front and Rear gates
- Single board no wiring loom
- Uses common through hole parts and modules
- Controlled via Bluetooth App on Android phone for configuring your requirements.
Download from google play store
- Battery connects directly to PCB via XT60/30
- Battery Type is auto detected. 4S-2S
- 3 Mode Select fire Switch
- Single Trigger Switch
- Rev Switch
- Magazine Sensor switch (Future program upgrade)
- Set to defaults by holding trigger while power up
- Via App: Display Battery Voltage
- Via App: Display Select Fire mode
- Via App: Display DPS
- Via App: Display Ammon Counter
- Via App: Trigger Button
- Via App: Game time countdown
- Via App: Burst Size slider
- Via App: Stage 1 Power slider
- Via App: Stage 2 Power slider
- Via App: Rev Ideal Spd slider
- Via App: Game Time slider
- To be added to app: Select fire Switch
- To be added to app: Rev Button
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- **NOTE: The App, does not work on a iPhone, it's very expensive to get a dev licence.**

Licensing

Non-commercial license is in effect. This license flows to derivative works.

Commercial use is allowed with the purchase of an RMB-Frankenboard from our Store at a ratio of one board = one commercial build.

Attributions

BLE Server examples

Based on Neil Kolban example for IDF: https://github.com/nkolban/esp32-snippets/blob/master/cpp_utils/tests/BLE%20Tests/SampleNotify.cpp

Ported to Arduino ESP32 by Evandro Copercini

Updated by chegewara and MoThunderz

(c) 2019-2021 Michael Ireland / Ireland Software / Airzone for example code used early on

License

RMB-Frankenboard

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Introduction

This is an advanced project, which requires some experience in electronics to complete.

You can purchase the Assembled and tested RMB – Frankenboard from our website site, we recommend this, to minimize frustrations in getting it going.

Please read, all documentation **before** starting this project.

Documentation:

- Frankenboard assembly, This file.
- RMB_ESP32_vnn.ino Arduino file. Read the header,.
- README.TXT, for helpful information.

FLASHING the ESP32 chip

Before starting this project, you need to flash the ESP32 chip, brand new out of the packet, while it has no wires or soldering done to it. This is very important, as you need to know the chip is working before you solder to it.

If you are not familiar with this procedure for the ESP32, we suggest you purchase the chip only before any other parts. Then flash the chip with the Arduino file.

NOTE you need a Micro USB-C with DATA SYNK cable. Not a phone charger cable.

Basic procedure: -

- Install Arduino IDE
- Load correct Board Library, XIAO ESP32 - S3, or XIAO ESP32 – C3
- Load required library's, into IDE for the code to compile. It will tell you what libarys it wants
- Get a successful compile
- Upload code (Flash chip)

After flashing, check for serial output in the IDE serial window, 115200 Baud, it will say BAT LOW.

If you get to this point, you have proven you have a 'working', RMB ESP32 chip.

Connect Blue Tooth aerial, load app to Android phone, Test app connection and operation.

If your having trouble flashing the ESP32, we recommend buying the assembled RMB-Frankinboard from our webpage.

BOM

1 x Micro USB-C Data Sync lead

1 x RMB-Frankenboard PCB 77 x 27mm 4 Layer

1 x U1 – Seeduino Xiao ESP32-S3. (C3 will work but is significantly slower)

1 x U2 - LR7843 Mosfet module

1 x U3 - MP2307DN 17x12mm Mini Buck, 5v (adjustable ones can vibrate and might be unreliable).

2 x U4, U5 – RPI-352 Optical Interrupter

2 x IN4007 Diodes D1, D2

2 x 22K Resistor R5, R8

2 x 330R Resistor R3, R6

1 x 47K Resistor R1

3 x 10K Resistor R2, R4 & R7

1 x XT60 Male plug

40 x 1 Pin Header Strip

The ESP32, Buck, and Mosfet module, mount via soldered pin headers.

The PCB is laid out to take standard size sockets and plugs.

Plugs and sockets introduce an unreliability in our finished product. I prefer to solder all wiring connections except the ESC signals. I use a 2-pin header for the ESC1 and ESC2 parts, as the ESC normally has a 2 or 3 pin socket on its lead.

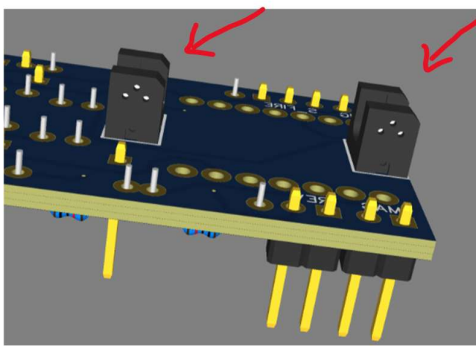
Assembly

At this point you should have a Firmware flashed RMB-ESP32

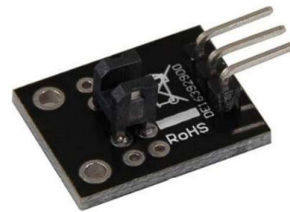
The order of putting parts on this PCB is important, as they go on both sides, and some solder points will not be accessible when other parts are mounted. All leads going through the PCB should be cut as short as possible.

Bottom side

Install the 2 x RPI-352 Optical Interrupter (Note the orientation). The bevelled corner faces the USB-end of the PCB.



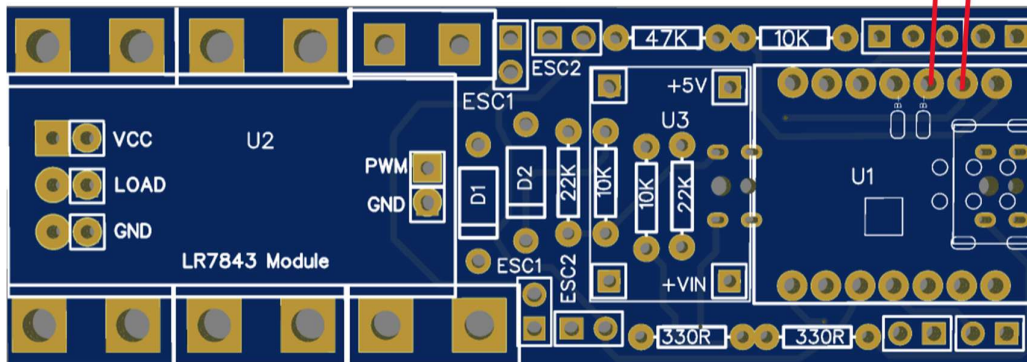
The component on the KY-010 interrupter module can be used, if you remove it



Top Side

Next install top side resistors and diodes (Note the direction of the diodes). See diagram below.

Red lines = Limit input test points



Mini Buck

This mounts on 4 header pins. Cut 4 single pin headers off pin header strip.

Place long end of pin into Frankenboard x 4.

Then place Buck module on top, hold assembly together and solder.

Cut off long legs from underside of Frankenboard.

First Power Up Test 1

Connect a XT60 to the Bat pads (GND on left POS on Right) and plug in Lipo Battery.

Using multi meter check between GND and 5v pin on ESP32 pad, should be 4-5v.

If Adjustable Buck, set output to ~5.7v as required to get 5.0v at ESP32 5v pin.

Measure between GND and PIN 2 (front limit gate) & PIN 3 (rear limit gate) (see red lines on pic above). Use a bit of tape to slide between interrupters. Voltage will be 0V open and 3V when interrupted

Disconnect battery.

ESP32

This mounts on 2 x 7way pin header strips. Cut 2 strips of 7 pins.

Insert long end of pin header strips into Frankenboard, ESP32 location.

Place ESP32 chip on top. It is keeping the pins in alignment for soldering to the Frankin PCB.

Hold assembly together and solder the pin header to Frankin PCB. Hint solder corner pin first then the rest of the pins. Then solder EPS32 to the pins.

Cut legs off from underside of PCB.

Install Pin Headers: -

Rev/Mag, 4 way

Trigger, 2 way

Select Fire Switch, 3 way

ESC1 and ESC2, 2 x 2 way left hand side, 2 x 2 way right hand side

Mosfet Module

This mounts on 5 pins. Cut 1 pair, and 3 single pins

Insert long end of pins into Frankin PCB

Place Mosfet module on top

Do NOT solder Mosfet Module on to any pins, yet.

Hold assembly together and solder pins to bottom side of Frankenboard, only

Remove Mosfet Module

Cut legs off from underside of PCB

Power Up Test 2

Do not connect any other wires yet. Do not connect USB to computer yet.

Start Arduino IDE, open serial window at 11520 baud and have USB-C cable ready, **DO NOT** plug into ESP32 yet.

Connect a Li-Po battery to the XT60 connector, and power up from Li-Po.

Now connect to USB-C and monitor serial window.

Do you have some serial output on screen that is readable.

Using Jumpers, on the pin headers, observe serial output and test:-

Trigger Switch

Mag Switch

Rev Switch

Select Fire Mode

Remove USB cable first then lipo Battery.

Final wiring

Insert Mosfet Module onto its pins and solder.

Connect ESC power cables and Solenoid.

ESC Configuration

The RMB - Frankenboard accommodates Single stage and Dual stage Flywheel/ESC configurations.

Single stage

Simply plug each ESC signal cable into the connections for ESC1 on LHS and ESC1 on RHS. Solder power wires into ESC Pwr connections.

Dual Stage

As above plus, ESC2 on LHS and ESC2 on RHS. Solder power wires into ESC Pwr connections.

The RMB – Frankinboard is tested for up to **XX amps**, 14 volts 4S lipo battery.

Final Power Up Test 3

Hold trigger down while you connect lipo battery. This clears and defaults the memory settings.

All going well you will hear the startup beeps of your ESC's.

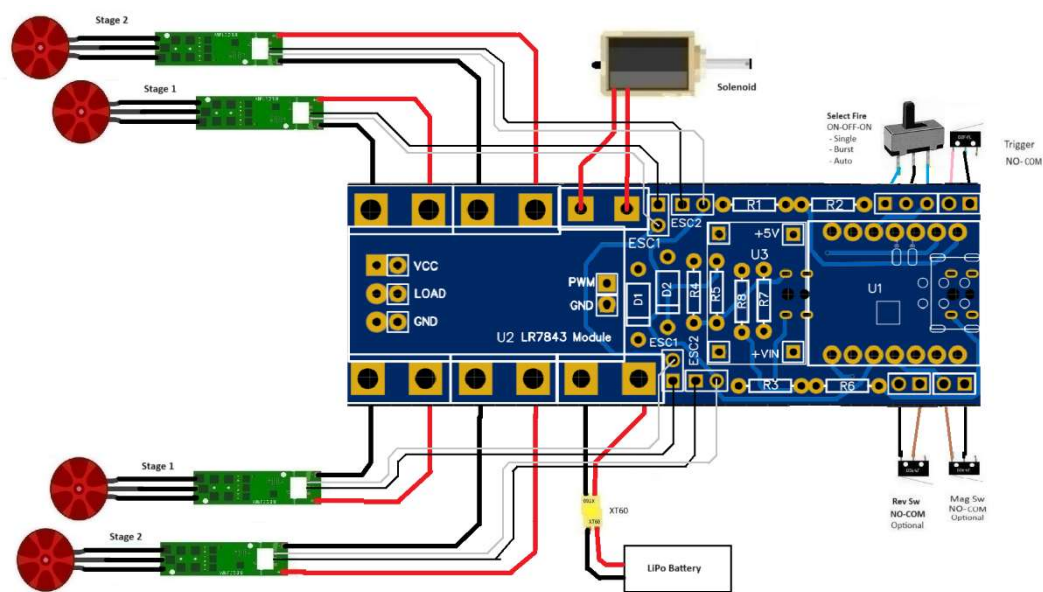
Using a jumper, on the select fire pins put it into single fire mode (jumper on furthest pins from USB).

Momentarily try a trigger test. Mosfet led should turn on.

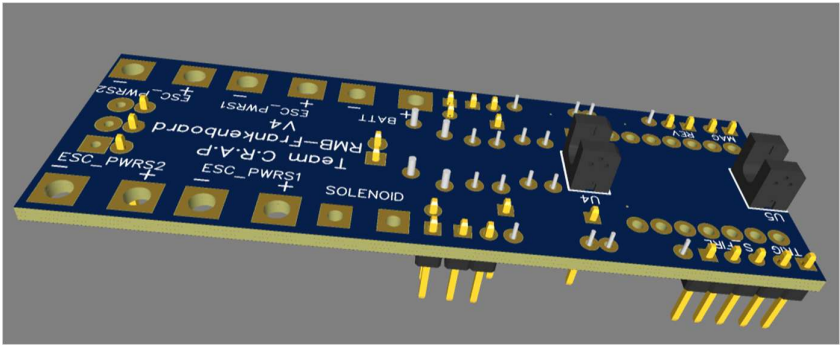
Wheels should spin and solenoid will fire.

If this doesn't happen, it would indicate a fault. Disconnect the battery and start searching for the fault.

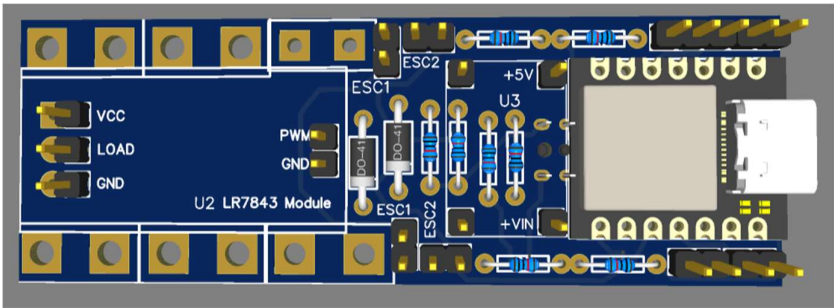
RMB - Frankenboard connection diagram



Note: Stage 2 is optional.

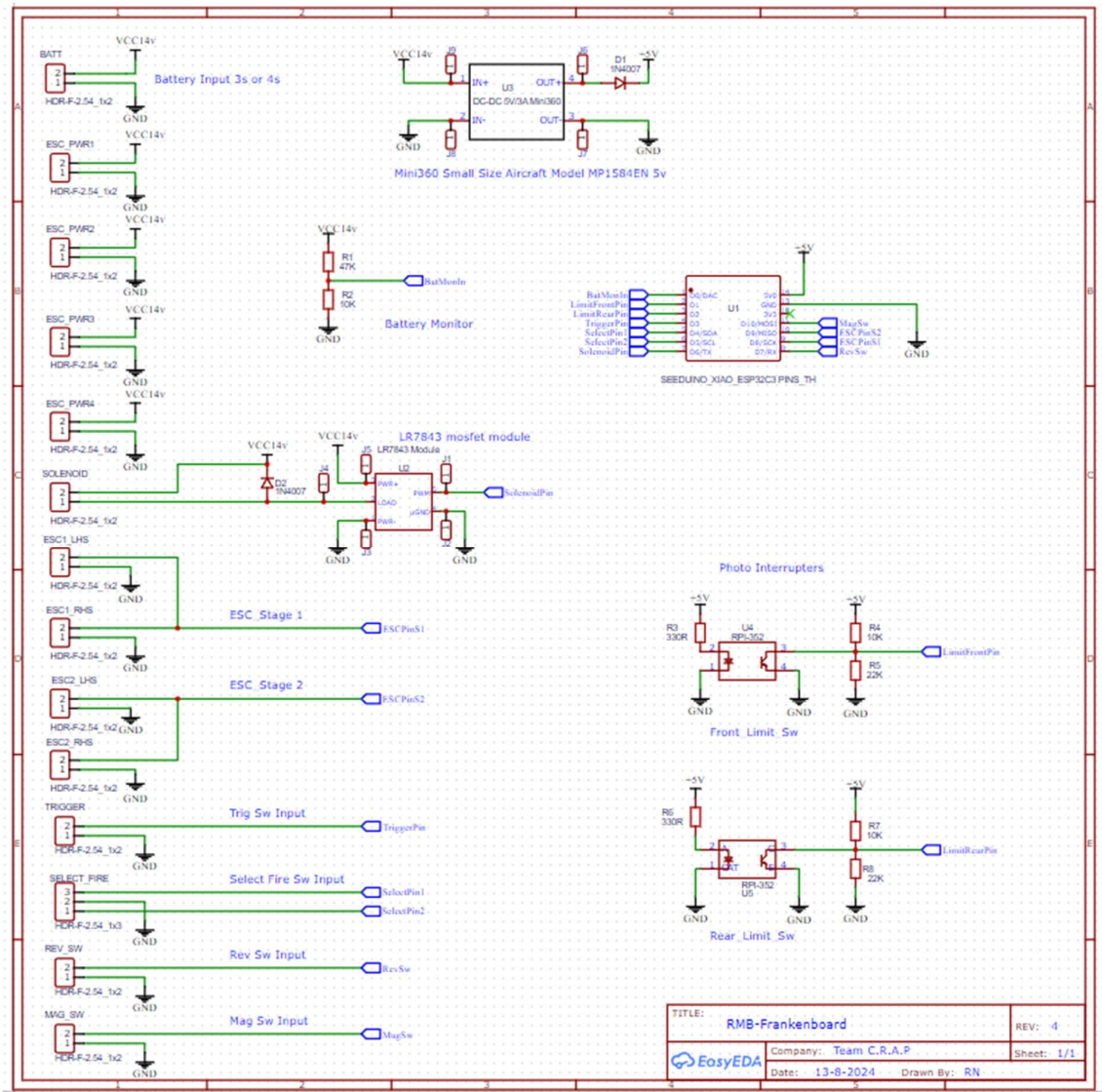


Bottom view.



Top View, without mosfet and Buck.

RMB-Frankenboard Schematic:



Flashing the ESP32 when fully wired

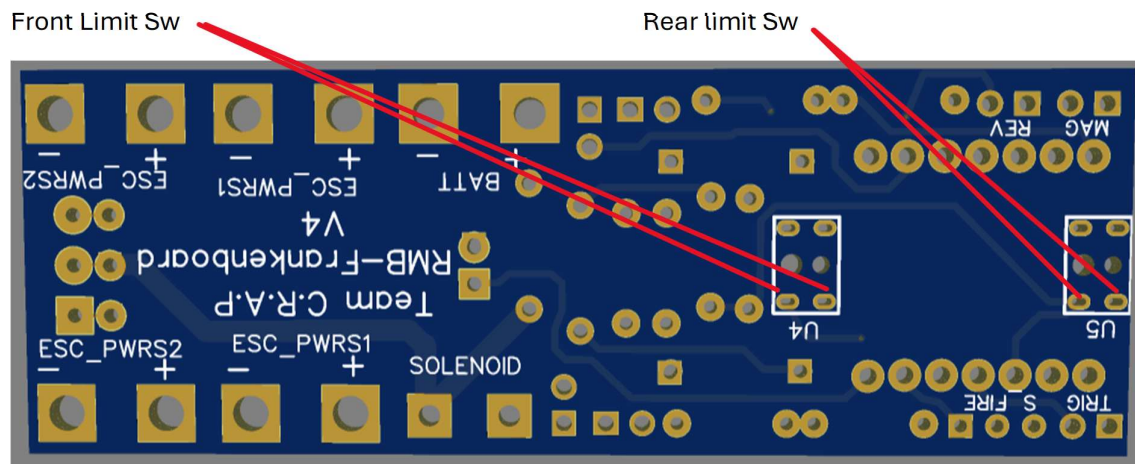
To ensure that the chip is in the correct boot mode please use this procedure:

The chip can appear to have stopped working if not done correctly.

1. Power up RMB-Frankenboard by Lipo battery.
2. Plug in USB Cable.
3. Start IDE and chk for serial coms is present. Pull the trigger.
4. Unplug the Lipo battery.
5. In IDE select download, program will compile and download.
6. Select serial coms and check output. First line on boot is the version number.
7. Unplug USB.
8. Plug in Lipo and test.

Alternative Limit Switch wiring

If you need to use physical limit switches. This can be accomplished by removing the Photo Interrupter gates, and connecting two wires to the pads indicated, to go to limit switch front and back.



Useful Links

XIAO ESP32-S3

<https://www.seeedstudio.com/XIAO-ESP32S3-p-5627.html>

XIAO ESP32 -C3

<https://www.seeedstudio.com/Seeed-XIAO-ESP32C3-p-5431.html>

Mini Buck, 5v (Do not get adjustable)

https://www.aliexpress.com/item/1005006486270630.html?spm=a2g0o.order_list.order_list_main.41.36121802kTN2ps&fbclid=IwY2xjawE24bFleHRuA2FlbQlxMAABHSmljFDdcVPMh2byogKuNxbBNBsBkU19hBsUyZnyoL12hfm4dXCDqF3iMw_aem_KDZTljSDMiwUsVOnOBZhA&gatewayAdapt=glo2deu

RPI-352 Optical Interrupter

https://www.aliexpress.com/item/1005005805132173.html?srcSns=sns_Messenger&spreadType=socialShare&bizType=ProductDetail&social_params=21610518952&aff_fcid=225def8a69d14ad0a98656d2fa166d47-1725070500008-06900-_mqhl1uU&tt=MG&fbclid=IwY2xjawE_W_NleHRuA2FlbQlxMAABHcbLkNNCJgfF3SDu6Kb4BOWr-iAxv--OOrpo1Etl7_rb7eyO5_8E4jNRHw_aem_Lnkp_anG8G9avq1Bz7w9yQ&aff_fsk=_mqhl1uU&aff_platform=default&sk=_mqhl1uU&aff_trace_key=225def8a69d14ad0a98656d2fa166d47-1725070500008-06900-_mqhl1uU&shareId=21610518952&businessType=ProductDetail&platform=AE&terminal_id=18184d2796b14b49999ce2660aa4400a&afSmartRedirect=y&gatewayAdapt=4itemAdapt

KY-010 Optical Interrupter can be used by removing part

https://www.aliexpress.com/item/1005006290314720.html?spm=a2g0o.productlist.main.3.3c218d3fzRX03W&algo_pvid=293b664d-da57-4e36-88b5-71214f39c9d5&algo_exp_id=293b664d-da57-4e36-88b5-71214f39c9d5-1&pdp_npi=4%40dis%21NZD%217.46%217.08%21%21%214.50%214.27%21%402101f93417245166007756847e4383%2112000036629387404%21sea%21NZ%211616880998%21X&curPageLogUid=0fXz8dX2Bgrf&utparam-url=scene%3Asearch%7Cquery_from%3A